

SATURDAY IN THE TRENCHES TASK

CHAPTER 2 · 15 PROGRAMMING CHALLENGES



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RULES & SUBMISSION

- 01 Do **not** use Artificial Intelligence in any form.
- 02 All tasks must be submitted into my **DM by 6:00 PM** this evening.
- 03 This is **not** a group work — individual submissions only.
- 04 Send a screenshot of your **pseudocode**, **working code**, and **terminal output**.

ENJOY YOUR WEEKEND!!!

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Questions 1–5

1. (ARITHMETIC)

Write an application that asks the user to enter two integers, obtains them from the user, and prints the cube of each number, the product of their cubes, and the remainder when the larger cube is divided by the smaller cube.

Hint: Use the techniques shown in the Addition program (Fig. 2.7).

2. (COMPARING INTEGERS)

Write an application that asks the user to enter one integer, obtains it from the user, and displays whether the number and its cube are greater than, equal to, or less than 500. Use the techniques shown in the Comparison program (Fig. 2.15).

3. (ARITHMETIC, SMALLEST AND LARGEST)

Write an application that inputs four integers from the user and displays the sum, integer average, product, smallest, and largest of the numbers. The average must be an integer — drop any decimal part.

Hint: Use only if statements and the techniques from Fig. 2.15.

4. (DISPLAYING SHAPES WITH ASTERISKS)

Write an application that uses only `System.out.println` statements to display the following three shapes made of asterisks (*): a right-angled triangle (4 rows), a hollow square (4×4), and a plus sign (+) that is 5 units tall and 5 units wide.

5. (LARGEST AND SMALLEST INTEGERS)

Write an application that reads six integers from the user and determines and prints the largest and the smallest integers in the group. Use only the programming techniques you have learned in this chapter.

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Questions 6 – 10

6. (DIVISIBLE BY 7)

Write an application that reads an integer from the user and determines and prints whether it is divisible by 7, not divisible by 7, or equal to zero. Also print whether the number is even or odd.

Hint: Use the remainder operator. A number is divisible by 7 if it's divided by 7 with a remainder of 0.

7. (MULTIPLES)

Write an application that reads two integers from the user, determines whether the first number squared is a multiple of the second number cubed, and prints the result with an appropriate message.

Hint: Use the remainder operator.

8. (DIAMETER, PERIMETER AND AREA OF A RECTANGLE)

Write an application that inputs from the user the length and width of a rectangle as integers and prints the perimeter, area, and the length of the diagonal using the value 1.41421 as an approximation for $\sqrt{2}$ where needed. Do not store results in variables — specify each calculation directly inside a `System.out.printf` statement.

Hint: For a rectangle with sides a and b , $\text{diagonal} = \sqrt{a^2 + b^2}$. You may approximate using only integer arithmetic and the multiplication operator.

9. (SEPARATING THE DIGITS IN AN INTEGER)

Write an application that inputs one four-digit integer from the user, separates the number into its individual digits, and prints each digit on its own line separated by three spaces. For example, if the user enters 5839, the program prints: 5 8 3 9.

Hint: Use both integer division and the remainder operator to extract each digit from left to right.

10. (NEGATIVE, POSITIVE AND ZERO VALUES)

Write an application that inputs six integers from the user and determines and prints the count of negative numbers entered, the count of positive numbers entered, and the count of zeros entered. Also print the sum of all positive numbers and the sum of all negative numbers separately.

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Questions 11 – 15

11. (TABLE OF SQUARES AND CUBES)

Using only the programming techniques you have learned in this chapter, write an application that calculates and prints the squares and cubes of the integers 1 through 5 in a neatly formatted table. Label the columns: `number`, `square`, `cube`. Use `printf` to align the values.

12. (ELECTRICITY BILL CALCULATOR)

Write an application that reads the number of kilowatt-hours (kWh) consumed by a household in a month as an integer. Calculate and print the electricity bill given: first 50 kWh at ₦25/kWh, next 100 kWh at ₦45/kWh, and anything above 150 kWh at ₦68/kWh. Use only the programming techniques from this chapter.

Hint: Use if statements and the remainder or subtraction operator to separate the usage tiers.

13. (SCHOOL FEES BREAKDOWN)

Write an application that reads a student's total school fees as an integer in naira. Print how many ₦1,000 notes, ₦500 notes, ₦200 notes, ₦100 notes, and ₦50 notes are needed to make up that exact amount (using the fewest notes possible). Assume the amount is a multiple of 50.

Hint: Use integer division to find how many of each note fit, then use the remainder operator to find what's left over before moving to the next denomination.

14. (BODY MASS INDEX CALCULATOR)

Write an application that reads a user's weight in kilograms and height in metres as integers, calculates and displays the BMI using the formula: $BMI = \text{weight} / (\text{height} \times \text{height})$. Display the BMI value and also print the category: "Underweight" (BMI < 18), "Normal" (BMI 18–24), "Overweight" (BMI 25–29), or "Obese" (BMI ≥ 30). Use only the techniques from this chapter.

15. (LAGOS TRAFFIC FINE CALCULATOR)

Write an application that reads a driver's speed in km/h as an integer. If the speed is 120 km/h or below, print "No violation." If the speed is 121–140 km/h, print "Warning — fine: ₦5,000." If the speed is 141–160 km/h, print "Serious violation — fine: ₦15,000." If above 160 km/h, print "Dangerous driving — fine: ₦50,000 and licence suspension." Use only if statements and the techniques from this chapter.

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